3.1. 加 $a_1+b_1\sqrt[3]{2}+c_1\sqrt[3]{4}+a_2+b_2\sqrt[3]{2}+c_2\sqrt[3]{4}=(a_1+a_2)+(b_1+b_2)\sqrt[3]{2}+(c_1+c_2)\sqrt[3]{4}$

减 $a_1+b_1\sqrt[3]{2}+c_1\sqrt[3]{4}-(a_2+b_2\sqrt[3]{2}+c_2\sqrt[3]{4})=(a_1-a_2)+(b_1-b_2)\sqrt[3]{2}+(c_1-c_2)\sqrt[3]{4}$

乘 $(a_1+b_1\sqrt[3]{2}+c_1\sqrt[3]{4})(a_2+b_2\sqrt[3]{2}+c_2\sqrt[3]{4})=(a_1a_2+2b_1c_2+2b_2c_1)+(a_1b_2+a_2b_1+2c_1c_2)\sqrt[3]{2}+(a_1c_2+a_2c_1+b_1b_2)\sqrt[3]{4}$

除 $\frac{a_1+b_1\sqrt[3]{2}+c_1\sqrt[3]{4}}{a_2+b_2\sqrt[3]{2}+c_2\sqrt[3]{4}}=\frac{(a_1+b_1\sqrt[3]{2}+c_1\sqrt[3]{4})[a_1^2-2b_1c_1+(2c_1^2-a_1b_1)\sqrt[3]{2}+(b_1^2-a_1c_1)\sqrt[3]{4}]}{(a_2+b_2\sqrt[3]{2}+c_2\sqrt[3]{4})[a_1^2-2b_1c_1+(2c_1^2-a_1b_1)\sqrt[3]{2}+(b_1^2-a_1c_1)\sqrt[3]{4}]}$

$=\frac{1}{a_1^3+2b_1^3+4c_1^3-6a_1b_1c_1}(a_1+b_1\sqrt[3]{2}+c_1\sqrt[3]{4})[a_1^2-2b_1c_1+(2c_1^2-a_1b_1)\sqrt[3]{2}+(b_1^2-a_1c_1)\sqrt[3]{4}]$

由乘知上式 $\in Q(\sqrt[3]{2})$,

故 $Q(\sqrt[3]{2})$ 满足封闭性

4、加 $a_1+b_1\sqrt{2}+c_1\sqrt{3}+d_1\sqrt{6}+a_2+b_2\sqrt{2}+c_2\sqrt{3}+d_2\sqrt{6}=(a_1+a_2)+(b_1+b_2)\sqrt{2}+(c_1+c_2)\sqrt{3}+(d_1+d_2)\sqrt{6}$

减 $a_1+b_1\sqrt{2}+c_1\sqrt{3}+d_1\sqrt{6}-(a_2+b_2\sqrt{2}+c_2\sqrt{3}+d\sqrt{6})=(a_1-a_2)+(b_1-b_2)\sqrt{2}+(c_1-c_2)\sqrt{3}+(d_1-d_3)\sqrt{6}$

乘 $(a_1+b_1\sqrt{2}+c_1\sqrt{3}+d_1\sqrt{6})(a_2+b_2\sqrt{2}+c_2\sqrt{3}+d_2\sqrt{6})=(a_1a_2+2b_1b_2+3c_1c_2+6d_1d_2)+(a_1b_2+a_2b_1+3c_2d_1+3c_1d_2)\sqrt{2}+(a_1c_2+a_2c_1+2b_1d_2+2b_2d_1)\sqrt{3}+(a_1d_2+a_2d_1+b_2c_1+b_1c_2)\sqrt{6}$

除 $\frac{a_1+b_1\sqrt{2}+kc_1\sqrt{3}+d_1\sqrt{6}}{a_2+b_2\sqrt{2}+c_2\sqrt{3}+d_2\sqrt{6}}=\frac{(a_1+b_1\sqrt{2}+c_1\sqrt{3}+d_1\sqrt{6})[a_2+b_2\sqrt{2}-\sqrt{3}(c_2+d_2\sqrt{2})]}{[a_2+b_2\sqrt{2}+\sqrt{3}(c_2+d_2\sqrt{2})][a_2+b_2\sqrt{2}-\sqrt{3}(c_2+d_2\sqrt{2})]}$ → 设为 $A\times B$

$=\frac{A\times B}{a_2^2+2b_2^2+2a_2b_2\sqrt{2}-3c_2^2-6d_2^2-6c_2d_2\sqrt{2}}$

$=\frac{A\times B\times C}{[a_2^2+2b_2^2-3c_2^2-6d_2^2+\sqrt{2}(2a_2b_2-6c_2d_2)][a_2^2+2b_2^2-3c_2^2-6d_2^2-\sqrt{2}(2a_2b_2-6c_2d_2)]}$

$=\frac{A\times B\times C}{(a_2^2+2b_2^2-3c_2^2-6d_2^2)^2-2(2a_2b_2-6c_2d_2)^2}$

由乘,知上式满足 $a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6}$

故 $Q(\sqrt{2},\sqrt{3})$ 满足封闭性

2.1 $\alpha+\beta+\gamma=(-2,0,0,0)$ $3\alpha-\beta+5\gamma=(0,-8,0,2)$

2.2 $3X=\gamma-\beta=(0,1,-2)$ $\therefore X=\frac{1}{3}(0,\frac{1}{3},-\frac{2}{3})$

3.1 (1) (3) (7) 是 (5) (9) 不是

(5) 对 $K(\alpha+\beta)=K\alpha+K\beta$ 左侧 $\rightarrow K(A\oplus B)=K(AB+BA)=KAB+KBA$

右侧 $\rightarrow KA\oplus KB=(K^2AB+K^2BA)=K^2AB+K^2BA$

两侧不一定相等

(9) 对加法交换律,取 $(2,2)$ $(1,1)$ $(2,2)+(1,1)=(3,1)\neq(1,1)+(2,2)=(3,-1)$

(1) (3) (7) 满足线性空间的各项条件

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2.7.1 $|A| = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ & \\ 1 & x_n \end{pmatrix} \times \begin{pmatrix} 1 & 1 & & 1 \\ y_1 & y_2 & \cdots & y_n \end{pmatrix} = 0$

2、由柯西毕内公式 原式表示 AA' 的一个主子式 (k阶介)

又∵ $AA' = I$, ∴ A 原式即 $\begin{pmatrix} 1 & \cdots & n \\ \cdots & \cdots & \cdots \end{pmatrix}$ 的行列式值，故值为

3、AB $\sum_{i=1}^{n-r+1} AB\begin{pmatrix} i & i+1 & \cdots & i+r-1 \\ i+1 & i+2 & \cdots & i+r \end{pmatrix} = \sum_{i=1}^{n-r+1} \sum_{1\le k_1<\cdots<k_r\le n} A\begin{pmatrix} i & i+1 & \cdots & i+r-1 \\ k_1 & k_2 & \cdots & k_r \end{pmatrix} B\begin{pmatrix} k_1 & k_2 & \cdots & k_r \\ i+1 & i+2 & \cdots & i+r \end{pmatrix}$

$\sum_{1\le i_1<\cdots<i_r\le n} AB\begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ \cdots & \cdots & \cdots & \cdots \end{pmatrix} = \sum_{1\le i_1<\cdots<i_r\le n} \sum_{1\le k_1<\cdots<k_r\le n} A\begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ k_1 & k_2 & \cdots & k_r \end{pmatrix} B\begin{pmatrix} k_1 & k_2 & \cdots & k_r \\ i_1 & i_2 & \cdots & i_r \end{pmatrix}$

$= \sum_{1\le k_1<\cdots<k_r\le n} \sum_{1\le i_1<\cdots<i_r\le n} B\begin{pmatrix} k_1 & k_2 & \cdots & k_r \\ i_1 & i_2 & \cdots & i_r \end{pmatrix} A\begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ k_1 & k_2 & \cdots & k_r \end{pmatrix}$

$= \sum_{1\le k_1<\cdots<k_r\le n} BA\begin{pmatrix} k_1 & k_2 & \cdots & k_r \\ k_1 & k_2 & \cdots & k_r \end{pmatrix}$

即 AB 和 BA 的所有 r 阶主子式相同

4、~~$|AA'| = \sum_{1\le i_1<\cdots<i_n\le n}$~~ $m>n$ 时 $|AA'| = |BB'| = |AB| = 0$

$m\le n$ 时 $|AA'| = \sum_{1\le i_1<\cdots<i_m\le n} A^2\begin{pmatrix} 1 & 2 & \cdots & m \\ i_1 & i_2 & \cdots & i_m \end{pmatrix}$

$|BB'| = \sum_{1\le i_1<\cdots<i_m\le n} B^2\begin{pmatrix} 1 & 2 & \cdots & m \\ i_1 & i_2 & \cdots & i_m \end{pmatrix}$

$|AB| = \sum_{1\le i_1<\cdots<i_m\le n} A\begin{pmatrix} 1 & 2 & \cdots & m \\ i_1 & i_2 & \cdots & i_m \end{pmatrix} B\begin{pmatrix} i_1 & i_2 & \cdots & i_m \\ 1 & 2 & \cdots & m \end{pmatrix}$

由 柯西不等式 可等原式成立

5、对 $z = m+ni$, $z\bar{z} = (m+ni)(m-ni) = m^2+n^2 \ge 0$

$A\bar{A}' = \sum_{1\le i_1<\cdots<i_m\le n} A\begin{pmatrix} 1 & 2 & \cdots & m \\ i_1 & i_2 & \cdots & i_m \end{pmatrix} \bar{A}\begin{pmatrix} 1 & 2 & \cdots & m \\ i_1 & i_2 & \cdots & i_m \end{pmatrix}$

易知为对称矩阵，$|B+\bar{B}| = \overline{|B|} = |B|$

∴ $A\begin{pmatrix} 1 & 2 & \cdots & m \\ i_1 & i_2 & \cdots & i_m \end{pmatrix} \bar{A}\begin{pmatrix} 1 & 2 & \cdots & m \\ i_1 & i_2 & \cdots & i_m \end{pmatrix}$ 的值非负

∴ $A\bar{A}'$ 的值非负

6、设 $A = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ b_1 & b_2 & \cdots & b_n \end{pmatrix}$

由5 $A\bar{A}' \ge 0$

又∵ $A\bar{A}' = \begin{vmatrix} \sum_{i=1}^n |a_i|^2 & \sum_{i=1}^n a_i\bar{b}_i \\ \sum_{i=1}^n \bar{a}_i b_i & \sum_{i=1}^n |b_i|^2 \end{vmatrix} = \left(\sum_{i=1}^n |a_i|^2\right)\left(\sum_{i=1}^n |b_i|^2\right) - \sum_{i=1}^n a_i\bar{b}_i \times \sum_{i=1}^n \bar{a}_i b_i \ge 0$

$=$

∴ $\left(\sum_{i=1}^n |a_i|^2\right)\left(\sum_{i=1}^n |b_i|^2\right) \ge \sum_{i=1}^n (a_i\bar{b}_i) \sum_{i=1}^n (\bar{a}_i b_i) = \sum_{i=1}^n (\bar{a}_i b_i) \sum_{i=1}^n (\bar{a}_i b_i) = \left|\sum_{i=1}^n a_i\bar{b}_i\right|^2$

3.4.2 设 $(1,1,a) = b(1,a,1) + c(a,1,1)$

∴ $\begin{cases} 1 = b+ac \\ 1 = ab+c \\ a = b+c \end{cases}$ ∴ $1 = b+bc+c^2 = b^2+bc+c$

∴ $(b-c)(b+c+1) = 0$

∴ $b=c$ 或 $b+c=a=-1$

∴ $\begin{cases} a=-2 \\ b=-1 \\ c=-1 \end{cases}$ 或 $\begin{cases} a=1 \\ b=\frac{1}{2} \\ c=\frac{1}{2} \end{cases}$ ∴ $a=1$ 或 $a=-2$

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4. 否，如 $\alpha_1=(1,1)$ $\alpha_2=(2,2)$ $\beta_1=(1,3)$ $\beta_2=(3,9)$

$\alpha_1+\alpha\beta_1=(2,4)$ $\alpha_2+\beta_2=(5,11)$

6. 否 如 $\alpha_1=(1,1)$ $\beta=(1,2)$ $\gamma=(2,3)$

8. 若这m个向量线性相关

则存在 $a_1, k_1 a_2 \cdots a_{m-1}$ 使

$\alpha_m=a_1\alpha_1+a_2\alpha_2+\cdots+a_{m-1}\alpha_{m-1}$

设 $\beta=b_1\alpha_1+b_2\alpha_2+\cdots+b_m\alpha_m$ ($b_1 b_2 \cdots b_m\neq 0$)

$\therefore \beta=(b_1+a_1b_m)\alpha_1+(b_2+a_2b_m)\alpha_2+\cdots+(b_{m-1}+a_{m-1}b_m)\alpha_{m-1}$ 矛盾

$\therefore$ ~~这mA个向量~~线性无关

10. 若 $A\alpha_1, A\alpha_2, \cdots A\alpha_m$ 线性相关，

则存在 $k_1 k_2 \cdots k_m\neq 0$ 使得

$k_1A\alpha_1+k_2A\alpha_2+\cdots+k_mA\alpha_m=0$

又$\because$A可逆

$\therefore k_1A^{-1}A\alpha_1+k_2A^{-1}A\alpha_2+\cdots+k_mA^{-1}A\alpha_m=0$ $A^{-1}\times 0$

$\therefore k_1\alpha_1+k_2\alpha_2+\cdots+k_m\alpha_m=0$ 矛盾

$\therefore A\alpha_1, A\alpha_2, \cdots A\alpha_m$ 线性无关

12. (1) 若 $\alpha_1, \alpha_2 \cdots \alpha_m$ 线性相关

则存在 $k_1, k_2 \cdots k_m$ 不全为0使 $k_1\alpha_1+k_2\alpha_2+\cdots+k_m\alpha_m=0$

$$\therefore \begin{cases} k_1a_{11}+k_2a_{21}+\cdots+k_ma_{m1}=0 \\ \vdots \\ k_1a_{1m}+k_2a_{2m}+\cdots+k_ma_{mm}=0 \end{cases}$$

$$\therefore \begin{cases} k_1a_{11}+k_2a_{21}+\cdots+k_ma_{m1}=0 \\ \vdots \\ k_1a_{1t}+k_2a_{2t}+\cdots+k_ma_{mt}=0 \end{cases}$$

$\therefore k_1\tilde{\alpha}_1+k_2\tilde{\alpha}_2+\cdots+k_m\tilde{\alpha}_m=0$

$\therefore \tilde{\alpha}_1, \tilde{\alpha}_2 \cdots \tilde{\alpha}_m$ 线性相关

(2) 若 $\alpha=k_1a_1+k_2a_2+\cdots+k_ma_m$

$$\therefore \begin{cases} \beta_1=k_1a_{11}+k_2a_{21}+\cdots+k_ma_{m1} \\ \vdots \\ a_m=k_1a_{1n}+k_2a_{2n}+\cdots+k_ma_{mn} \end{cases} \quad \therefore \begin{cases} \tilde{\alpha}_1=k_1a_{11}+k_2a_{21}+\cdots+k_ma_{m1} \\ \vdots \\ a_t=k_1a_{1t}+k_2a_{2t}+\cdots+k_ma_{mt} \end{cases}$$

$\therefore \tilde{\alpha}=k_1\tilde{a}_1+k_2\tilde{a}_2+\cdots+k_m\tilde{a}_m$ $\therefore \alpha$ 是 $a_1, a_2 \cdots a_m$ 的线性组合

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